

MUHAMMAD TALHA SHAIKH

Junior Full-Stack & AI Engineer

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PROFESSIONAL SUMMARY

High-velocity **Junior Full-Stack & AI Engineer** (~1.5 years intensive building experience) trained through **GIAIC (Governor Sindh Initiative for GenAI & Web3)**, with a proven track record of architecting and deploying scalable web systems, agentic AI pipelines, and multi-tenant platforms solo. Shipped **Botaura** (162+ route multi-tenant RAG SaaS with Meta WhatsApp Tech Provider accreditation), **FlowCreator OS** (autonomous AI video directing engine with locked facial geometry via Google Gemini SDK), **ShieldTools** (zero-server client-side privacy & e-commerce economics platform with PWA), and **Volvelo** (48-route luxury European marketplace with Clerk RBAC sync). Dedicated to rapid skill acquisition, clean system design, and immediate engineering impact.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript (ES6+), Python, Next.js 16 (App Router, Turbopack, Server Actions), React 19, FastAPI, Node.js, HTML5 Canvas API, jsPDF

AI & Vector Retrieval (RAG): Google Gemini 2.5/3.6 Flash SDK, Reference Image Anchoring, Temporal Choreography, OpenAI SDK, OpenAI Agent SDK, LangChain, pgvector, BM25 Hybrid Retrieval, Structured JSON (Zod)

Databases & ORM: Neon Serverless PostgreSQL, Prisma ORM 6, Drizzle ORM, Supabase, MongoDB

Cloud, Edge & DevOps: Vercel (Edge & Cron), Docker, Cloudflare R2, AWS S3, Git & GitHub Actions, Sentry, Hugging Face Spaces

Integrations & APIs: WhatsApp Cloud API (Meta Tech Provider), Green-API, CallMeBot, Clerk Auth (Core 3 / RBAC Sync), Stripe Checkout, Resend, Sanity Headless CMS

Architecture & Best Practices: Zero-Knowledge Client Architecture, Multi-Tenant Row-Level Security, Progressive Web Apps (PWA), Answer Engine Optimization (AEO/JSON-LD)

FEATURED SYSTEMS & ENGINEERING PROJECTS

Botaura — Multi-Tenant RAG AI Chatbot & WhatsApp Commerce SaaS [Flagship]

Founder & Solo Engineer

Live: <https://botaura.app> · Case Study: talhawebsite.xyz/projects/botaura

- Architected and deployed a multi-tenant SaaS platform spanning **162+ production routes**, enabling SMBs to train custom AI chatbots on proprietary knowledge bases and automate WhatsApp sales.
- Implemented Hybrid RAG pipeline combining OpenAI embeddings with pgvector cosine similarity and BM25 search for bilingual English/Roman-Urdu queries with automated confidence thresholds.
- Registered as Meta WhatsApp Tech Provider: built centralized token routing by phone_number_id, orchestrating real-time webhooks, templates, and ad attribution across multiple merchants.
- Enforced strict server-side tenant isolation with signed JWTs, UUID cascading, and hashed API keys, preventing cross-tenant data leakage by design.

Tech: Next.js 16, TypeScript, FastAPI, Python (asyncpg), PostgreSQL (Neon), pgvector, Cloudflare R2, Clerk Auth, Resend, Meta Graph API.

FlowCreator OS — Autonomous AI Video Directing & CreatorOps Operating System

Lead System Architect

Case Study: talhawebsite.xyz/projects/flow-creator-os

- Engineered Autonomous AI Director using Google Gemini 2.5/3.6 Flash SDK to transform creative premises into camera-sequenced video production bundles for Google Veo 2 and Sora.
- Designed Reference Image Anchoring injecting master biometric tokens to eliminate diffusion face-drift, and temporal match-cuts with continuous room-tone.
- Built an Infinite 8-Season Continuity Escalation Engine with 1-click season transitions and automated PKT time-zoned WhatsApp reminders via Green-API & CallMeBot.

Tech: Next.js 16, React 19, Tailwind CSS v4, Google Gemini Flash, Neon Postgres, Prisma ORM 6, Clerk Auth, Vercel Cron, WhatsApp API.

ShieldTools — Zero-Server Digital Privacy & COD E-Commerce Economics Platform

Lead Architect & Creator

Live: <https://tools.talhaweb.xyz/> · Case Study: talhaweb.xyz/projects/shield-tools

- Built 100% in-browser zero-knowledge watermarking studio (DocShield™) using HTML5 Canvas API and jsPDF to watermark sensitive national IDs (CNIC/Passports) directly in RAM with \$0 backend cost.
- Engineered EcomShield™ COD unit economics engine isolating courier reverse shipping penalties, packaging waste, and burned CAC on COD returns with automated WhatsApp summaries.
- Implemented installable Progressive Web App (PWA) with early beforeinstallprompt event capture, achieving sub-second load times across 22 statically prerendered routes.

Tech: Next.js 16 (App Router + SSG), TypeScript, Tailwind CSS v4, HTML5 Canvas API, jsPDF, PWA Service Worker.

Volvelo — Luxury Multi-Tenant European Dropshipping Marketplace & Atelier Hub

Full-Stack Architect

Case Study: talhaweb.xyz/projects/volvelo

- Architected 48-route platform separating customer storefront, merchant atelier portal (/portal), and super-admin hub (/admin) with strict tenant partitioning on Neon PostgreSQL via Prisma ORM.
- Built bidirectional Clerk Cloud RBAC synchronization with Neon PostgreSQL, ensuring instant permission propagation to user session claims without re-login.
- Engineered dynamic client-side multi-currency engine (EUR, USD, GBP, PKR) with Zustand persistence and hydration guards avoiding React 19 SSR hydration mismatches.

Tech: Next.js 16, React 19, TypeScript, Tailwind CSS v4, Neon Postgres, Prisma ORM, Clerk Core 3, Zustand.

GetF4F — P2P TikTok Social Growth & Heuristic Proof Verification Exchange

Full-Stack Engineer

Live: <https://f4f-tiktok.vercel.app/> · Case Study: talhaweb.xyz/projects/getf4f-tiktok

- Designed an asynchronous 24-hour delayed matching engine to mimic natural creator discovery and prevent mutual-follow algorithmic penalties on TikTok.
- Implemented heuristic screenshot validation analyzing profile UI signals, button states, and handles backed by a 100-point Trust Score and 5-day dispute grace period.
- Developed double-entry credit ledger, streak gamification mechanics, and emergency admin kill switches using Drizzle ORM on Neon PostgreSQL.

Tech: Next.js 16, React 19, TypeScript, Tailwind CSS v4, Neon Postgres, Drizzle ORM, Clerk Auth, AWS S3, Resend, PWA.

EDUCATION & CREDENTIALS

Cloud Native Applied Generative AI & Full-Stack Engineering	— Governor Sindh Initiative (GIAIC)	Ongoing
Meta WhatsApp Tech Provider Accreditation	— Verified Cloud API Integration Architecture	Verified
Intensive Project-Based Systems Development	— Real-world production full-stack & AI software	2022 – Present